一、Completions (5 items, 2 points each)

1. $\int_0^a \sqrt{a^2-x^2}dx \quad (a>0)=$_____.

2. Suppose $f(x)$ is continuous on $[a,b]$ and $f(x)\neq 0$, then $\lim\limits_{n\to\infty}\sqrt[n]{\int_a^b |f(x)|dx}=$_____.

3. The average value of $f(x)=e^x$ on $[0,1]$ is _____.

4. The domain of $\Gamma(x)=\int_0^{+\infty} t^{x-1}e^{-t}dt$ is _____.

5. Assume $f(x)$ is continuous on $[0,+\infty)$ and $\int_0^{x^2(x+1)} f(t)\,dt = x$, then $f(2)$ =_____.

二、Choices (5 items, 2 points each)

Score	

6. If $F(x)=\int_x^{x+\pi} e^{\cos 2t}dt$, then $F(x)$ is ( )

A. a positive constant. B. a negative constant.
C. not a constant. D. 0.

7. Let $M=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\sin x}{1+x^2}\cos^4 x dx$, $N=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\left(\sin^3 x+\cos^4 x\right)dx$, $P=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\left(x^2\sin^3 x-\cos^4 x\right)dx$, then ( )

A. $N<P<M$. B. $M<P<N$.
C. $N<M<P$. D. $P<M<N$.

8. The convergent improper integral is ( )

A. $\int_0^{+\infty} e^{-\sqrt{x}}dx$. B. $\int_0^{+\infty}\frac{1}{x^2}dx$. C. $\int_1^{+\infty}\frac{1}{\sqrt{x}}dx$. D. $\int_0^1\frac{1}{x^2}dx$.

9. Suppose $f(x)$ has second derivative on $[0,1]$, and $f(x)\geq 0$, $f''(x)>0$, then ( )

A. $\int_0^1 f(x)\,dx<\frac{f(0)+f(1)}{2}$. B. $\int_0^1 f(x)\,dx>\frac{f(0)+f(1)}{2}$.
C. $\int_0^1 f(x)\,dx=0$. D. $\frac{f(0)+f(1)}{2}<f\left(\frac{1}{2}\right)$.

10. Let $F(x)$ be an antiderivative of $f(x)$ on an interval I, the correct one in the following propositions is ( )

19. Find the positive constants a and b, such that $\lim_{x\to 0}\frac{1}{bx-\sin x}\int_0^x \frac{t^2}{\sqrt{a+t^2}}\,dt=1$.

Score	

五、Proof Problem（8 points）

20. Let $f(x)$ be continuous on $[0,1]$,

(1) Prove that $\int_0^{\pi} xf(\sin x)\,dx=\frac{\pi}{2}\int_0^{\pi} f(\sin x)\,dx$;

(2) Calculate $\int_0^{\pi}\frac{x\sin x}{1+\cos^2 x}\,dx$.